# Hecksher Ohlin Theory and RCA

## International Economics - T6

Prof. Kalyan Kolukuluri

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IIM Visakhapatnam

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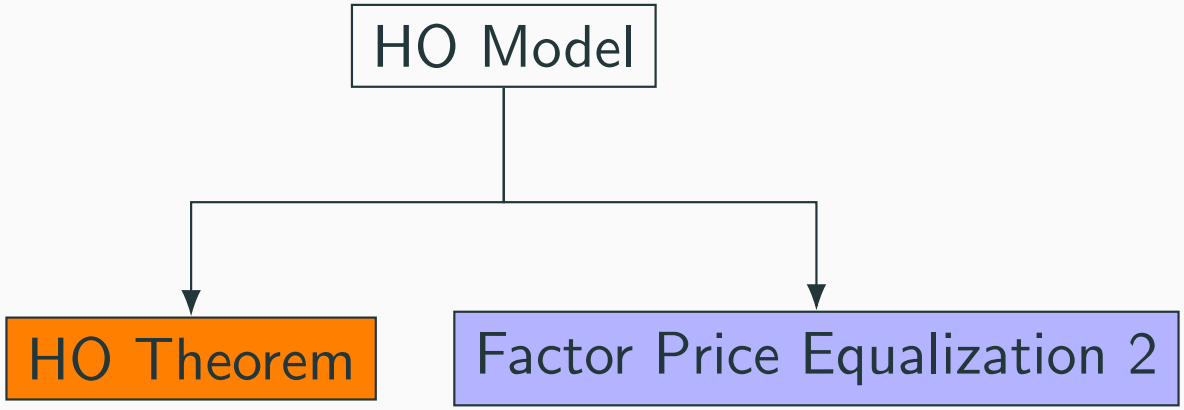
## Learning Goals

- Explain how comparative advantage is based on differences in factor endowments across nations
- Explain how trade affects relative factor prices within and across nations

# Factor Endowment, Intensity and Trade (Hecksher-Ohlin Model)

# Heckscher Ohlin Model



# H-O Theorem

**The H-O Theorem**

A nation will export the commodity whose production requires the intensive use of the nation's relatively abundant and cheap factor and import the commodity whose production requires the intensive use of the nation's relatively scarce and expensive factor.

## Factor Intensity

- In a two-commodity, two-factor world, commodity Y is capital intensive if the capital-labor ratio (K/L) used in the production of Y is greater than K/L used in the production of X.
- It is not the absolute amount of capital and labor used in the production of X and Y, but the amount of capital per unit of labor that determines capital intensity.

## Factor Abundance (Physical Units)

- Nation 2 is capital abundant if the ratio of the total amount of capital to the total amount of labor (TK/TL) available in Nation 2 is greater than that in Nation 1.
- It is not the absolute amount of capital and labor available in each nation, but the ratio of the total amount of capital to the total amount of labor.

## Factor Abundance (Factor Prices)

- Nation 2 is capital abundant if the ratio of the rental price of capital to the price of labor time (PK/PL) is lower in Nation 2 than in Nation 1.
- PK/PL = r/w. $r$ is rental rate and $w$ is wage rate
- It is not the absolute level of r that determines whether a nation is K-abundant, but r/w.

# Factor Price Equalization

## Factor Price Equalization

International trade causes wages ($w$) to rise in Nation 1 (the low-wage nation) and fall in Nation 2 (the high-wage nation), reducing the pre-trade difference in wages between nations.
Similarly, trade causes the return on capital ($r$) to fall in Nation 1 (the capital-expensive nation) and rise in Nation 2 (the capital-cheap nation), reducing the pre-trade difference in return on capital between nations.

# Revealed Comparative Advantage - Knowing Oneself

# Revealed Comparative Advantage (Balassa (1965))

## Revealed Comparative Advantage

RCA: A country's share of world exports in one commodity divided by its share of total world exports

- RCA $> 1$ is an advantage.
- $0 <$ RCA $< \infty$

$$\text{RCA} = \frac{\dfrac{X_{ij}}{\sum_i (X_{ij})}}{\dfrac{\sum_j (X_{ij})}{\sum_i \sum_j X_{ij}}} = \frac{\dfrac{\text{Export of TV's by India}}{\text{Total World TV's Export}}}{\dfrac{\text{India's total exports}}{\text{World's Total Exports}}},$$

- X is the exports from a country.
- $i$ is the index for the good; $j$ is the index for the country

# RCA India (Top 10) - 2023

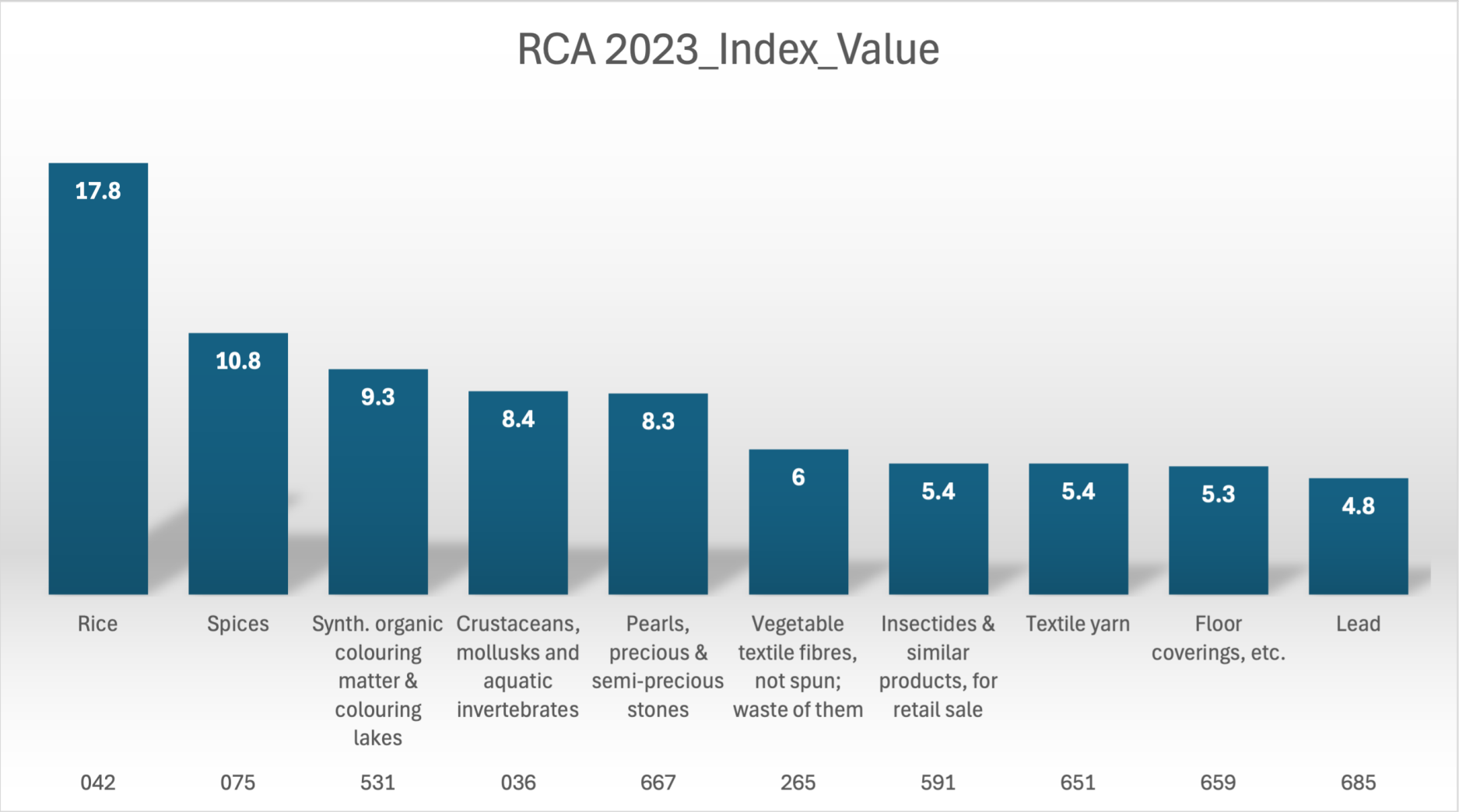



**Figure 1:** India RCA (Top 10) - 2023

## RCA Radar Plots

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## Q & A

THANK YOU !!!

Q & A

# The End